

Commercial Resupply Services (CRS) are a series of flights awarded by NASA for the delivery of cargo and supplies to the International Space Station (ISS) on commercially operated spacecraft.

The first phase of CRS contracts (CRS-1) were signed in 2008 and awarded \$1.6 billion to SpaceX for twelve Dragon 1 and \$1.9 billion to Orbital Sciences for eight Cygnus flights, covering deliveries to 2016. The first operational resupply missions were flown by SpaceX in 2012 (CRS SpX-1) and Orbital in 2014 (CRS Orb-1). In 2015, NASA extended CRS-1 to twenty flights for SpaceX and twelve flights for Orbital ATK.

A second phase of contracts (CRS-2) was solicited in 2014. CRS-2 contracts were awarded in January 2016 to Orbital ATK's continued use of Cygnus, Sierra Nevada Corporation's new Dream Chaser, and SpaceX's new Dragon 2, for cargo transport flights beginning in 2019 and expected to last through 2024.

COTS demo flights

NASA has been directed to pursue commercial spaceflight options since at least 1984, with the Commercial Space Launch Act of 1984 and Launch Services Purchase Act of 1990. By the 2000s funding was authorized for the Commercial Orbital Transportation Services (COTS) program, followed by the Commercial Crew Development program. The Antares and Falcon 9 launch vehicles and Cygnus and Dragon cargo spacecraft were developed using Space Act Agreements under the COTS program.

The first flight contracted by NASA, COTS Demo Flight 1, took place on December 8, 2010, demonstrating a Dragon capsule's ability to remain in orbit, receive and respond to ground commands, and communicate with NASA's Tracking and Data Relay Satellite System. On August 15, 2011, SpaceX announced that NASA had combined the objectives of the COTS Demo Flight 2 and following Flight 3 into a single mission. The rescoped COTS Demo Flight 2 successfully launched on May 22, 2012, delivering cargo to the ISS. The spacecraft reentered

on May 31, landed in the Pacific Ocean, and was recovered, completing CRS certification requirements.

Orbital Sciences first launched the Antares rocket from the Mid-Atlantic Regional Spaceport on April 21, 2013, with a test payload. Orbital Sciences completed the Cygnus Orb-D1 demonstration flight on September 29, 2013, and the operational Cygnus CRS Orb-1 was launched January 9, 2014.

CRS-1 selection process

On December 23, 2008, NASA announced the initial awarding of cargo contracts - twelve flights to SpaceX and eight flights to Orbital Sciences Corporation. PlanetSpace, which was not selected, submitted a protest to the Government Accountability Office. On April 22, 2009, the GAO publicly released its decision to deny the protest, allowing the program to continue.

CRS-1 flights

Transport flights began under Commercial Resupply Services phase 1 (CRS-1) in 2012:

Cargo Dragon flights

CRS SpX-1: October 8, 2012

CRS SpX-2: March 1, 2013

CRS SpX-3: April 18, 2014

CRS SpX-4: September 21, 2014.

CRS SpX-5: January 10, 2015

CRS SpX-6: April 14, 2015

CRS SpX-7: attempted on June 28, 2015. Launch failure 139 seconds after lift-off, IDA-1 destroyed. Investigation traced the accident to the failure of a strut inside the second stage's liquid-oxygen tank. NASA concluded that the most probable cause of the strut failure was a

design error: instead of using a stainless-steel eye bolt made of aerospace-grade material, SpaceX chose an industrial-grade material without adequate screening and testing and overlooked the recommended safety margin.

CRS SpX-8: April 8, 2016

CRS SpX-9: July 18, 2016

CRS SpX-10: February 19, 2017

CRS SpX-11: June 3, 2017

CRS SpX-12: August 14, 2017

CRS SpX-13: December 15, 2017

CRS SpX-14: April 2, 2018

CRS SpX-15: June 29, 2018

CRS SpX-16: December 5, 2018

CRS SpX-17: May 4, 2019

CRS SpX-18: July 25, 2019

CRS SpX-19: December 5, 2019

CRS SpX-20: March 7, 2020

Cygnus flights

CRS Orb-1: January 9, 2014

CRS Orb-2: July 13, 2014

CRS Orb-3: October 28, 2014 - launch failure, food and care packages for the crew, parts, experiments, and the Arkyd-3 Flight Test (Non-optical) Satellite from Planetary Resources lost.

Following the failure, the Antares rocket was upgraded to the 230 series which used newly built RD-181 first-stage engines to provide greater payload performance and increased reliability. The next two spacecraft were launched on the Atlas V. With the switch to more powerful launch vehicles and the introduction of a larger Enhanced Cygnus, enabled Orbital

ATK to cover their initial CRS contracted payload obligation by OA-7.

CRS OA-4: December 6, 2015 - Atlas V, first Enhanced Cygnus

CRS OA-6: March 23, 2016 - Atlas V

CRS OA-5: October 17, 2016 - Antares 230

CRS OA-7: April 18, 2017 - Atlas V

During August 2015, Orbital ATK disclosed that they had received an extension of the resupply program for four extra missions. These flights enable NASA to cover ISS resupply needs until CRS-2 begins.

CRS OA-8E: November 12, 2017

CRS OA-9E: May 21, 2018

CRS NG-10: November 17, 2018

CRS NG-11: April 17, 2019

CRS-2 selection process

NASA began a formal process to initiate Phase 2 of the Commercial Resupply Services, or CRS-2, in early 2014. Later that year, an "Industry Day" was held in Houston, with seven high-level requirements disclosed to interested parties.

Requirements

The contracts were expected to include a variety of requirements:

delivery of approximately 14,000 to 17,000 kg (31,000 to 37,000 lb) per year 55 to 70 m³ (1,900 to 2,500 cu ft) of pressurized cargo in four or five transport trips

delivery of 24–30 powered lockers per year, requiring continuous power of up to 120 watts at

28 volts, cooling, and two-way communications

delivery of approximately 1,500 to 4,000 kg (3,300 to 8,800 lb) per year of unpressurized cargo, consisting of 3 to 8 items, each item requiring continuous power of up to 250 watts at 28 volts, cooling, and two-way communications

return/disposal of approximately 14,000 to 17,000 kg (31,000 to 37,000 lb) per year 55 to 70 m³ (1,900 to 2,500 cu ft) of pressurized cargo

disposal of 1,500 to 4,000 kg (3,300 to 8,800 lb) per year of unpressurized cargo, consisting of 3 to 8 items

various ground support services

Proposals

CRS-1 contractors Orbital Sciences and SpaceX each submitted CRS-2 proposals, joined by Sierra Nevada, Boeing, and Lockheed Martin.

SNC's proposal would use a cargo version of its Dream Chaser crew vehicle, the 'Dream Chaser Cargo System'. The proposed cargo Dream Chaser included an additional expendable cargo module for uplift and trash disposal. Downmass would only be provided via the Dream Chaser spaceplane itself. Boeing's proposal likewise used a cargo version of its CST-100 crew vehicle.

Lockheed Martin proposed a new cargo spacecraft called Jupiter, derived from the designs of the NASA's MAVEN and Juno spacecraft. It would have included a robotic arm based on Canadarm technology and a 4.4-meter (14 ft) diameter cargo transport module called Exoliner based on the Automated Transfer Vehicle, to be jointly developed with Thales Alenia Space.

Awards

When NASA issued the Commercial Resupply Services phase 2 (CRS-2) request for proposal (RFP) in September 2014, it received interest from five companies: Lockheed Martin, Boeing,

Orbital ATK, Sierra Nevada, and SpaceX. NASA made a competitive range determination to remove Boeing and Lockheed Martin.

Three companies were awarded contracts on January 14, 2016. Sierra Nevada Corporation's Dream Chaser, the SpaceX Dragon 2, and Orbital ATK Cygnus were selected, each for a minimum of six launches. The maximum potential value of all the contracts was indicated to be \$14 billion, but the minimum value is considerably less. CRS-2 launches commenced in 2019 and were to extend to at least 2024. NASA officials explained that selecting three companies rather than two for CRS-2 increases cargo capabilities and ensures more redundancy in the event of a contractor failure or schedule delay.

Three more CRS-2 missions for Dragon 2 covering up to CRS-29 were announced in December 2020, and six more were awarded in 2022 covering up to CRS-35. The contracts were then extended to 2030.

Northrop Grumman having purchased Orbital ATK were awarded 2 more Cygnus flights in 2020 and 6 more in 2022 covering up to NG-35 to provide resupply services to the station through 2026. The contracts were then extended to 2030.

The CRS-2 flights commenced in November 2019 with the launch of Cygnus NG-12 mission.

Inside-cargo is typically transported to and from the space station in "the form factor of single Cargo Transfer Bag Equivalent (CTBE) [which is the] unit for size of bag used to transport cargo from visiting vehicles, such as SpaceX Dragon, Northrop Grumman Cygnus, or JAXA H-II Transfer Vehicle (HTV). The bags are sized at 48 cm × 41 cm × 23 cm (19 in × 16.25 in × 9 in) and limited in transport mass to 27 kg (60 lb) each. CTBE units are also used to price, and charge, commercial users of US Orbital Segment stowage space.

CRS-2 flights

Cygnus flights

CRS NG-12: November 2, 2019

CRS NG-13: February 15, 2020

CRS NG-14: October 3, 2020

CRS NG-15: February 20, 2021

CRS NG-16: August 10, 2021

CRS NG-17: February 19, 2022

CRS NG-18: November 7, 2022

CRS NG-19: August 2, 2023

As a result Russia's invasion of Ukraine, Northrop Grumman was left with only two remaining Antares 230+ launch vehicles which were used for the CRS NG-18 and CRS NG-19 missions. Northrop Grumman initially acquired three flights from SpaceX with the Falcon 9 rocket while a replacement first stage and its engine are developed for its Antares 330 rocket. They added additional Falcon 9 launches as Antares was further delayed.

CRS NG-20: January 30, 2024 - Falcon 9

CRS NG-21: August 4, 2024 - Falcon 9

CRS NG-22: TBD (indefinitely delayed after cargo module damaged during shipment)

CRS NG-23: September 14, 2025 - Falcon 9, first Cygnus XL

CRS NG-24: April 8, 2026 - Falcon 9

CRS NG-25: 2026 (planned) - Falcon 9

Northrop Grumman plans to launch further missions using the new Antares 300 series (Antares 330) rockets with booster stage and engines developed by Firefly Aerospace.

Cargo Dragon flights

CRS SpX-21: December 6, 2020

CRS SpX-22: June 3, 2021

CRS SpX-23: August 29, 2021

CRS SpX-24: December 21, 2021

CRS SpX-25: July 15, 2022

CRS SpX-26: November 26, 2022

CRS SpX-27: March 14, 2023

CRS SpX-28: June 5, 2023

CRS SpX-29: November 10, 2023

CRS SpX-30: March 21, 2024

CRS SpX-31: November 5, 2024

CRS SpX-32: April 21, 2025

CRS SpX-33: August 24, 2025

CRS SpX-34: May 15, 2026

CRS SpX-35: August 2026 (planned)

Cargo Dream Chaser flights

SSC Demo-1: late 2026 (planned)

See also

Commercial Crew Development (CCDev) – development of crew vehicles

Commercial Lunar Payload Services

Comparison of space station cargo vehicles

Multi-Purpose Logistics Module

Uncrewed spaceflights to the International Space Station

Notes

References

External links

NASA Commercial Crew and Cargo webpage

SpaceX (Space Exploration Technologies Corporation)

OSC (Orbital Sciences Corporation)

CRS contracts and Space Act agreements